# Lec-13

Friday, 9 October 2026 3:46 pm

## Cubic Spline Interpolation[1]

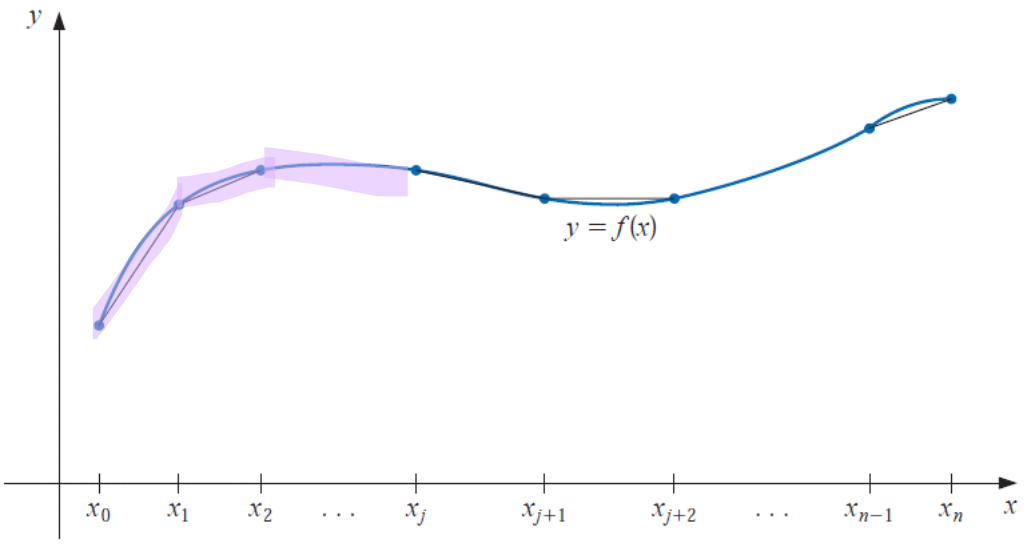
The previous sections concerned the approximation of arbitrary functions on closed intervals using a single polynomial. However, high-degree polynomials can oscillate erratically, that is, a minor fluctuation over a small portion of the interval can induce large fluctuations over the entire range. We will see a good example of this in Figure 3.14 at the end of this section.

An alternative approach is to divide the approximation interval into a collection of subintervals and construct a (generally) different approximating polynomial on each subinterval. This is called **piecewise-polynomial approximation**.

### Piecewise-Polynomial Approximation

The simplest piecewise-polynomial approximation is **piecewise-linear** interpolation, which consists of joining a set of data points

$$\{(x_0, f(x_0)), (x_1, f(x_1)), \ldots, (x_n, f(x_n))\}$$



Isaac Jacob Schoenberg (1903–1990) developed his work on splines during World War II while on leave from the University of Pennsylvania to work at the Army's Ballistic Research Laboratory in Aberdeen, Maryland. His original work involved numerical procedures for solving differential equations. The much broader application of splines to the areas of data fitting and computer-aided geometric design became evident with the widespread availability of computers in the 1960s.

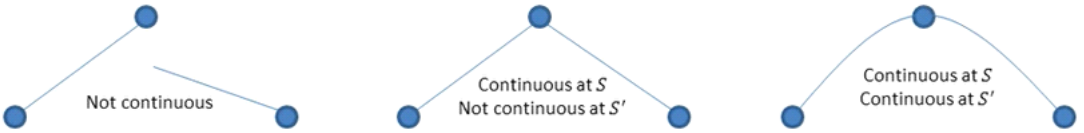
## Cubic Splines

The most common piecewise-polynomial approximation uses cubic polynomials between each successive pair of nodes and is called **cubic spline interpolation**. A general cubic polynomial involves four constants, so there is sufficient flexibility in the cubic spline procedure to ensure that the interpolant is not only continuously differentiable on the interval, but also has a continuous second derivative. The construction of the cubic spline does not, however, assume that the derivatives of the interpolant agree with those of the function it is approximating, even at the nodes. (See Figure 3.8.)

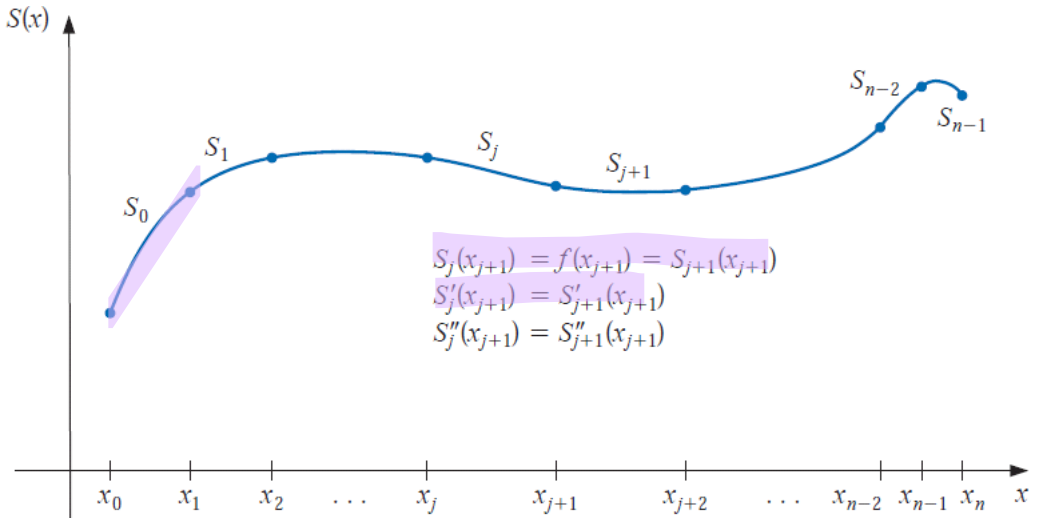
The root of the word "spline" is the same as that of splint. It was originally a small strip of wood that could be used to join two boards. Later the word was used to refer to a long flexible strip, generally of metal, that could be used to draw continuous smooth curves by forcing the strip to pass through specified points and tracing along the curve.

# What's Special About Splines?

- Cubic Splines are function approximations that are continuous at merging points.
- They also have continuous first and second derivatives where they join.




- Natural Cubic Spline is a cubic spline that has second derivative equal to zero at end points.
- Unlike Lagrange Polynomials, each point in the line does not change the entire line .




$$S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3,$$

**Definition 3.10** Given a function $f$ defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \cdots < x_n = b$, a **cubic spline interpolant** $S$ for $f$ is a function that satisfies the following conditions:

(a) $S(x)$ is a cubic polynomial, denoted $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, 1, \ldots, n-1$;

(b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each $j = 0, 1, \ldots, n-1$;

(c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$; (*Implied by* (b).)

(d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$;

(e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$;

(f) One of the following sets of boundary conditions is satisfied:

(i) $S''(x_0) = S''(x_n) = 0$ (**natural** *(or free)* **boundary**);

(ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (**clamped boundary**). ■

A natural spline has no conditions imposed for the direction at its endpoints, so the curve takes the shape of a straight line after it passes through the interpolation points nearest its endpoints. The name derives from the fact that this is the natural shape a flexible strip assumes if forced to pass through specified interpolation points with no additional constraints. (See Figure 3.9.)

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(i) $S''(x_0) = S''(x_n) = 0$ (**natural *(or free)* boundary**);

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**Example 1** Construct a natural cubic spline that passes through the points $(1, 2)$, $(2, 3)$, and $(3, 5)$.

***Solution*** This spline consists of two cubics. The first for the interval $[1, 2]$, denoted

$$S_0(x) = a_0 + b_0(x - 1) + c_0(x - 1)^2 + d_0(x - 1)^3,$$

and the other for $[2, 3]$, denoted

$$S_1(x) = a_1 + b_1(x - 2) + c_1(x - 2)^2 + d_1(x - 2)^3.$$

There are 8 constants to be determined, which requires 8 conditions. Four conditions come from the fact that the splines must agree with the data at the nodes. Hence

$$2 = f(1) = a_0, \quad 3 = f(2) = a_0 + b_0 + c_0 + d_0, \quad 3 = f(2) = a_1, \quad \text{and}$$
$$5 = f(3) = a_1 + b_1 + c_1 + d_1.$$

Two more come from the fact that $S_0'(2) = S_1'(2)$ and $S_0''(2) = S_1''(2)$. These are

$$S_0'(2) = S_1'(2): \quad b_0 + 2c_0 + 3d_0 = b_1 \quad \text{and} \quad S_0''(2) = S_1''(2): \quad 2c_0 + 6d_0 = 2c_1$$

The final two come from the natural boundary conditions:

$$S_0''(1) = 0: \quad 2c_0 = 0 \quad \text{and} \quad S_1''(3) = 0: \quad 2c_1 + 6d_1 = 0.$$

Solving this system of equations gives the spline

$$S(x) = \begin{cases} 2 + \frac{3}{4}(x - 1) + \frac{1}{4}(x - 1)^3, & \text{for } x \in [1, 2] \\ 3 + \frac{3}{2}(x - 2) + \frac{3}{4}(x - 2)^2 - \frac{1}{4}(x - 2)^3, & \text{for } x \in [2, 3] \end{cases}$$

■

$b_0 + d_0 = 1$ ; $b_1 + c_1 + d_1 = 2$ ; $b_0 + 3d_0 = b_1$ ; $3d_0 = c_1$ ; $2c_1 + 6d_1 = 0$

$b_0 \quad d_0 \quad b_1 \quad c_1 \quad d_1$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & -1 & 0 & 0 & 0 \\ 0 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 6 & 0 \end{array}\right] \Rightarrow$$

$b_0 = \frac{3}{4}$, $d_0 = \frac{1}{4}$, $b_1 = \frac{3}{2}$, $c_1 = \frac{3}{4}$

$d_1 = -\frac{1}{4}$

$q_0 =$

# How do you calculate a cubic spline?

- Given $(x_1, y_1)$, $(x_2, y_2)$, and $(x_3, y_3)$
- $ax_2^3 + bx_2^2 + cx_2 + d = y_2$
- $ex_2^3 + fx_2^2 + gx_2 + h = y_2$
- $3ax_2^2 + 2bx_2 + c = 3ex_2^2 + 2fx_2 + g$
- $6ax_2 + 2b = 6ex_2 + 2f$
- $6ax_1 + 2b = 0$
- $6ex_3 + 2f = 0$

- Example:

Given:

$S_1(x) = 2x^3 - 3x^2 + 3x - 4, 0 \le x \le 1$ and
$S_2(x) = x^3 + ax^2 + bx + c, 1 \le x \le 2$

Find $a$, $b$, and $c$ that make this a cubic spline.

$S_1(x) = 2x^3 - 3x^2 + 3x - 4$, $0 \le x \le 1$

- $S_1(1) = 2(1)^3 - 3(1)^2 + 3(1) - 4 = -2$
- $S_1'(1) = 6(1)^2 - 6(1) + 3 = 3$
- $S_1''(1) = 12(1) - 6 = 6$

$S_2(x) = x^3 + ax^2 + bx + c$, $1 \le x \le 2$

- $S_2(1) = (1)^3 + a(1)^2 + b(1) + c = 1 + a + b + c = -2$
- $S_2'(1) = 3(1)^2 + 2a(1) + b = 3 + 2a + b = 3$
- $S_2''(1) = 6(1) + 2a = 6$

$$1 + a + b + c = -2$$
$$3 + 2a + b = 3$$
$$6 + 2a = 6$$
$$a = 0, b = 0, c = -3$$

1. Determine the natural cubic spline $S$ that interpolates the data $f(0) = 0$, $f(1) = 1$, and $f(2) = 2$.
2. Determine the clamped cubic spline $s$ that interpolates the data $f(0) = 0$, $f(1) = 1$, $f(2) = 2$ and satisfies $s'(0) = s'(2) = 1$.